

Finite Dimensional Vector Spaces (2015–2021)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS

34

TOTAL MARKS

55 M

TEST DURATION

102 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2015–20)

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	18	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	5	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	11	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q60 • ID: JAM_2021_Q60)

NAT

+2 M

Neg: Nil (0)

Q. 60 Let $A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$. Consider the linear map T_A from the real vector space $M_4(\mathbb{R})$ to itself defined by $T_A(X) = AX - XA$, for all $X \in M_4(\mathbb{R})$. The dimension of the range of T_A is ____.

Question 2 (JAM 2021 • Q46 • ID: JAM_2021_Q46)

NAT

+1 M

Neg: Nil (0)

Q. 46 Let V be the real vector space of all continuous functions $f : [0, 2] \rightarrow \mathbb{R}$ such that the restriction of f to the interval $[0, 1]$ is a polynomial of degree less than or equal to 2, the restriction of f to the interval $[1, 2]$ is a polynomial of degree less than or equal to 3 and $f(0) = 0$. Then the dimension of V is equal to ____.

MA

14 / 17

Question 3 (JAM 2021 • Q39 • ID: JAM_2021_Q39)

MSQ

+2 M

Neg: Nil (0)

Q. 39 Let V be a finite dimensional vector space and $T : V \rightarrow V$ be a linear transformation. Let $\mathcal{R}(T)$ denote the range of T and $\mathcal{N}(T)$ denote the null space $\{v \in V : Tv = 0\}$ of T . If $\text{rank}(T) = \text{rank}(T^2)$, then which of the following is/are necessarily true?

(A) $\mathcal{N}(T) = \mathcal{N}(T^2)$.(B) $\mathcal{R}(T) = \mathcal{R}(T^2)$.(C) $\mathcal{N}(T) \cap \mathcal{R}(T) = \{0\}$.(D) $\mathcal{N}(T) = \{0\}$.

Question 4 (JAM 2021 • Q29 • ID: JAM_2021_Q29)

MCQ

+2 M

Neg: -0.67

JAM 2021

MATHEMATICS - MA

Q. 29 Let $n \geq 2$ be an integer. Let $A : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be the linear transformation defined by

$$A(z_1, z_2, \dots, z_n) = (z_n, z_1, z_2, \dots, z_{n-1}).$$

Which one of the following statements is true for every $n \geq 2$?

- (A) A is nilpotent. (B) All eigenvalues of A are of modulus 1.
 (C) Every eigenvalue of A is either 0 or 1. (D) A is singular.

Question 5 (JAM 2021 • Q11 • ID: JAM_2021_Q11)

MCQ

+2 M

Neg: -0.67

Q. 11 – Q. 30 carry two marks each.

Q. 11 Let $M_n(\mathbb{R})$ be the real vector space of all $n \times n$ matrices with real entries, $n \geq 2$.

Let $A \in M_n(\mathbb{R})$. Consider the subspace W of $M_n(\mathbb{R})$ spanned by $\{I_n, A, A^2, \dots\}$. Then the dimension of W over $\mathbb{R}$ is necessarily

- (A) ∞ . (B) n^2 . (C) n . (D) at most n .

Question 6 (JAM 2021 • Q9 • ID: JAM_2021_Q9)

MCQ

+1 M

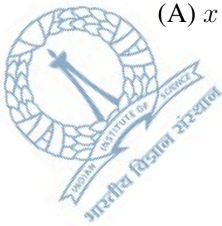
Neg: -0.33

Q. 9 For an integer $k \geq 0$, let P_k denote the vector space of all real polynomials in one variable of degree less than or equal to k . Define a linear transformation $T : P_2 \rightarrow P_3$ by

$$Tf(x) = f''(x) + xf(x).$$

Which one of the following polynomials is not in the range of T ?

- (A) $x + x^2$ (B) $x^2 + x^3 + 2$ (C) $x + x^3 + 2$ (D) $x + 1$



MA

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Question 7 (JAM 2020 • Q59 • ID: JAM_2020_Q59)

NAT

+2 M

Neg: Nil (0)

Q. 59 Let $T : \mathbb{R}^7 \rightarrow \mathbb{R}^7$ be a linear transformation with $\text{Nullity}(T) = 2$. Then, the minimum possible value for $\text{Rank}(T^2)$ is _____.

Question 8 (JAM 2020 • Q49 • ID: JAM_2020_Q49)

NAT

+1 M

Neg: Nil (0)

Q. 49 Consider the real vector space $P_{2020} = \left\{ \sum_{i=0}^n a_i x^i : a_i \in \mathbb{R} \text{ and } 0 \leq n \leq 2020 \right\}$. Let W be the subspace given by

$$W = \left\{ \sum_{i=0}^n a_i x^i \in P_{2020} : a_i = 0 \text{ for all odd } i \right\}.$$

Then, the dimension of W is _____.

Question 9 (JAM 2020 • Q34 • ID: JAM_2020_Q34)

MSQ

+2 M

Neg: Nil (0)

JAM 2020

MATHEMATICS - MA

Q. 34 Let V be a non-zero vector space over a field F . Let $S \subset V$ be a non-empty set. Consider the following properties of S :

(I) For any vector space W over F , any map $f : S \rightarrow W$ extends to a linear map from V to W .

(II) For any vector space W over F and any two linear maps $f, g : V \rightarrow W$ satisfying $f(s) = g(s)$ for all $s \in S$, we have $f(v) = g(v)$ for all $v \in V$.

(III) S is linearly independent.

(IV) The span of S is V .

Which of the following statement(s) is /are true?

(A) (I) implies (IV)

(B) (I) implies (III)

(C) (II) implies (III)

(D) (II) implies (IV)

Question 10 (JAM 2020 • Q23 • ID: JAM_2020_Q23)

MCQ

+2 M

Neg: -0.67

Q. 23 Let M be a 4×3 real matrix and let $\{e_1, e_2, e_3\}$ be the standard basis of $\mathbb{R}^3$. Which of the following is true?

- (A) If $\text{rank}(M) = 1$, then $\{Me_1, Me_2\}$ is a linearly independent set
- (B) If $\text{rank}(M) = 2$, then $\{Me_1, Me_2\}$ is a linearly independent set
- (C) If $\text{rank}(M) = 2$, then $\{Me_1, Me_3\}$ is a linearly independent set
- (D) If $\text{rank}(M) = 3$, then $\{Me_1, Me_3\}$ is a linearly independent set

Question 11 (JAM 2020 • Q15 • ID: JAM_2020_Q15)

MCQ

+2 M

Neg: -0.67

Q. 15 Let $f(x, y) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0, y = 0 \\ y^2 \sin \frac{1}{y}, & y \neq 0, x = 0 \\ 0, & x = y = 0. \end{cases}$

Which of the following is true at $(0, 0)$?

- (A) f is not continuous
- (B) $\frac{\partial f}{\partial x}$ is continuous but $\frac{\partial f}{\partial y}$ is not continuous
- (C) f is not differentiable
- (D) f is differentiable but both $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are not continuous

MA

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Question 12 (JAM 2020 • Q4 • ID: JAM_2020_Q4)

MCQ

+1 M

Neg: -0.33

Q. 4 Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function. If $f(x, y) = g(y) + xg'(y)$, then

(A) $\frac{\partial f}{\partial x} + y \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial y}$

(B) $\frac{\partial f}{\partial y} + y \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial x}$

(C) $\frac{\partial f}{\partial x} + x \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial y}$

(D) $\frac{\partial f}{\partial y} + x \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial x}$

MA

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Question 13 (JAM 2019 • Q42 • ID: JAM_2019_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42 Let W_1 and W_2 be subspaces of the real vector space $\mathbb{R}^{100}$ defined by

$$W_1 = \{ (x_1, x_2, \dots, x_{100}) : x_i = 0 \text{ if } i \text{ is divisible by } 4 \},$$

$$W_2 = \{ (x_1, x_2, \dots, x_{100}) : x_i = 0 \text{ if } i \text{ is divisible by } 5 \}.$$

Then the dimension of $W_1 \cap W_2$ is _____

Question 14 (JAM 2019 • Q33 • ID: JAM_2019_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33 Let S and T be linear transformations from a finite dimensional vector space V to itself such that $S(T(v)) = 0$ for all $v \in V$. Then

(A) $\text{rank}(T) \geq \text{nullity}(S)$

(B) $\text{rank}(S) \geq \text{nullity}(T)$

(C) $\text{rank}(T) \leq \text{nullity}(S)$

(D) $\text{rank}(S) \leq \text{nullity}(T)$

Question 15 (JAM 2019 • Q20 • ID: JAM_2019_Q20)

MCQ

+2 M

Neg: -0.67

Q.20 Let S be the family of orthogonal trajectories of the family of curves

$$2x^2 + y^2 = k, \text{ for } k \in \mathbb{R} \text{ and } k > 0.$$

If $\mathcal{C} \in S$ and $\mathcal{C}$ passes through the point $(1, 2)$, then $\mathcal{C}$ also passes through

- (A) $(4, -\sqrt{2})$ (B) $(2, -4)$ (C) $(2, 2\sqrt{2})$ (D) $(4, 2\sqrt{2})$

Question 16 (JAM 2019 • Q4 • ID: JAM_2019_Q4)

MCQ

+1 M

Neg: -0.33

Q.4 Let $\{a_n\}_{n=0}^{\infty}$ and $\{b_n\}_{n=0}^{\infty}$ be sequences of positive real numbers such that $na_n < b_n < n^2a_n$ for all $n \geq 2$. If the radius of convergence of the power series $\sum_{n=0}^{\infty} a_n x^n$ is 4, then the power series $\sum_{n=0}^{\infty} b_n x^n$

- (A) converges for all x with $|x| < 2$
 (B) converges for all x with $|x| > 2$
 (C) does not converge for any x with $|x| > 2$
 (D) does not converge for any x with $|x| < 2$

Question 17 (JAM 2019 • Q1 • ID: JAM_2019_Q1)

MCQ

+1 M

Neg: -0.33

Q. 1 – Q.10 carry one mark each.

Q.1 Let $a_1 = b_1 = 0$, and for each $n \geq 2$, let a_n and b_n be real numbers given by

$$a_n = \sum_{m=2}^n \frac{(-1)^m m}{(\log(m))^m} \text{ and } b_n = \sum_{m=2}^n \frac{1}{(\log(m))^m}.$$

Then which one of the following is TRUE about the sequences $\{a_n\}$ and $\{b_n\}$?

- (A) Both $\{a_n\}$ and $\{b_n\}$ are divergent
 (B) $\{a_n\}$ is convergent and $\{b_n\}$ is divergent
 (C) $\{a_n\}$ is divergent and $\{b_n\}$ is convergent
 (D) Both $\{a_n\}$ and $\{b_n\}$ are convergent

Question 18 (JAM 2018 • Q57 • ID: JAM_2018_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57 Suppose $Q \in M_{3 \times 3}(\mathbb{R})$ is a matrix of rank 2. Let $T: M_{3 \times 3}(\mathbb{R}) \rightarrow M_{3 \times 3}(\mathbb{R})$ be the linear transformation defined by $T(P) = QP$. Then the rank of T is _____

Question 19 (JAM 2018 • Q49 • ID: JAM_2018_Q49)

NAT

+1 M

Neg: Nil (0)

Q.49 Let W_1 be the real vector space of all 5×2 matrices such that the sum of the entries in each row is zero. Let W_2 be the real vector space of all 5×2 matrices such that the sum of the entries in each column is zero. Then the dimension of the space $W_1 \cap W_2$ is _____

Question 20 (JAM 2018 • Q26 • ID: JAM_2018_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 Let H be the quotient group $\mathbb{Q}/\mathbb{Z}$. Consider the following statements.

- I. Every cyclic subgroup of H is finite.
- II. Every finite cyclic group is isomorphic to a subgroup of H .

Which one of the following holds?

- (A) I is TRUE but II is FALSE
- (B) II is TRUE but I is FALSE
- (C) both I and II are TRUE
- (D) neither I nor II is TRUE

Question 21 (JAM 2018 • Q16 • ID: JAM_2018_Q16)

MCQ

+2 M

Neg: -0.67

Q.16 For $x \in \mathbb{R}$, let $f(x) = \begin{cases} x^3 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$. Then which one of the following is FALSE?

- (A) $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 0$
- (B) $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 0$
- (C) $\frac{f(x)}{x^2}$ has infinitely many maxima and minima on the interval $(0,1)$
- (D) $\frac{f(x)}{x^4}$ is continuous at $x = 0$ but not differentiable at $x = 0$

Question 22 (JAM 2018 • Q6 • ID: JAM_2018_Q6)

MCQ

+1 M

Neg: -0.33

Q.6 In $\mathbb{R}^3$, the cosine of the acute angle between the surfaces $x^2 + y^2 + z^2 - 9 = 0$ and $z - x^2 - y^2 + 3 = 0$ at the point $(2, 1, 2)$ is

(A) $\frac{8}{5\sqrt{21}}$

(B) $\frac{10}{5\sqrt{21}}$

(C) $\frac{8}{3\sqrt{21}}$

(D) $\frac{10}{3\sqrt{21}}$

Question 23 (JAM 2017 • Q46 • ID: JAM_2017_Q46)

NAT

+1 M

Neg: Nil (0)

Q.46 Let $v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. Let M be the matrix whose columns are $v_1, v_2, 2v_1 - v_2, v_1 + 2v_2$ in that order. Then the number of linearly independent solutions of the homogeneous system of linear equations $Mx = 0$ is _____

Question 24 (JAM 2017 • Q22 • ID: JAM_2017_Q22)

MCQ

+2 M

Neg: -0.67

Q.22 Let $\mathcal{P}_3$ denote the real vector space of all polynomials with real coefficients of degree at most 3. Consider the map $T: \mathcal{P}_3 \rightarrow \mathcal{P}_3$ given by $T(p(x)) = p''(x) + p(x)$. Then

(A) T is neither one-one nor onto
 (B) T is both one-one and onto
 (C) T is one-one but not onto
 (D) T is onto but not one-one

Question 25 (JAM 2016 • Q59 • ID: JAM_2016_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59 Let $A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 5 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ -1 & 0 \\ 3 & 1 \end{bmatrix}$, $N(A)$ the null space of A and $R(B)$ the range space of B . Then the dimension of $N(A) \cap R(B)$ over $\mathbb{R}$ is _____

Question 26 (JAM 2016 • Q39 • ID: JAM_2016_Q39)

MSQ

+2 M

Neg: Nil (0)

Q.39 Consider the set $V = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \mid \alpha x + \beta y + z = \gamma, \alpha, \beta, \gamma \in \mathbb{R} \right\}$. For which of the following choice(s) the set V becomes a two dimensional subspace of $\mathbb{R}^3$ over $\mathbb{R}$?

- (A) $\alpha = 0, \beta = 1, \gamma = 0$
 (B) $\alpha = 0, \beta = 1, \gamma = 1$
 (C) $\alpha = 1, \beta = 0, \gamma = 0$
 (D) $\alpha = 1, \beta = 1, \gamma = 0$

Question 27 (JAM 2016 • Q27 • ID: JAM_2016_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 Let $M = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & \alpha \\ 2 & -\alpha & 0 \end{bmatrix}$, $\alpha \in \mathbb{R} \setminus \{0\}$ and $\mathbf{b}$ a non-zero vector such that $M\mathbf{x} = \mathbf{b}$ for some $\mathbf{x} \in \mathbb{R}^3$. Then the value of $\mathbf{x}^T \mathbf{b}$ is

- (A) $-\alpha$ (B) α (C) 0 (D) 1

Question 28 (JAM 2016 • Q17 • ID: JAM_2016_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{xy^2}{x+y} & \text{if } x+y \neq 0 \\ 0 & \text{if } x+y = 0 \end{cases}.$$

Then the value of $\left(\frac{\partial^2 f}{\partial x \partial y} + \frac{\partial^2 f}{\partial y \partial x} \right)$ at the point $(0, 0)$ is

- (A) 0 (B) 1 (C) 2 (D) 4

Question 29 (JAM 2016 • Q2 • ID: JAM_2016_Q2)

MCQ

+1 M

Neg: -0.33

Q.2 Let P be the vector space (over $\mathbb{R}$) of all polynomials of degree ≤ 3 with real coefficients. Consider the linear transformation $T: P \rightarrow P$ defined by

$$T(a_0 + a_1 x + a_2 x^2 + a_3 x^3) = a_3 + a_2 x + a_1 x^2 + a_0 x^3.$$

Then the matrix representation M of T with respect to the ordered basis $\{1, x, x^2, x^3\}$ satisfies

- (A) $M^2 + I_4 = 0$ (B) $M^2 - I_4 = 0$
 (C) $M - I_4 = 0$ (D) $M + I_4 = 0$

Question 30 (JAM 2015 • Q54 • ID: JAM_2015_Q54)

NAT

+2 M

Neg: Nil (0)

Q.14 Let $M_2(\mathbb{R})$ be the vector space of 2×2 real matrices. Let V be a subspace of $M_2(\mathbb{R})$ defined by

$$V = \left\{ A \in M_2(\mathbb{R}) : A \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} A \right\}$$

Then the dimension of V is _____

MA

15/16

Question 31 (JAM 2015 • Q48 • ID: JAM_2015_Q48)

NAT

+1 M

Neg: Nil (0)

Q.8

If the set $\left\{ \begin{bmatrix} x & -x \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ x & x \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} \right\}$ is linearly dependent in the vector space of all 2×2 matrices with real entries, then x is equal to _____

Question 32 (JAM 2015 • Q35 • ID: JAM_2015_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.5

Let V be the set of 2×2 matrices $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ with complex entries such that $a_{11} + a_{22} = 0$. Let W be the set of matrices in V with $a_{12} + \overline{a_{21}} = 0$. Then, under usual matrix addition and scalar multiplication, which of the following is (are) true?

- (A) V is a vector space over $\mathbb{C}$
- (B) W is a vector space over $\mathbb{C}$
- (C) V is a vector space over $\mathbb{R}$
- (D) W is a vector space over $\mathbb{R}$

Question 33 (JAM 2015 • Q17 • ID: JAM_2015_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 The limit

is equal to

$$\lim_{x \rightarrow 0^+} \frac{1}{\sin^2 x} \int_{\frac{x}{2}}^x \sin^{-1} t \, dt$$

(A) 0

(B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{3}{8}$

Question 34 (JAM 2015 • Q4 • ID: JAM_2015_Q4)

MCQ

+1 M

Neg: -0.33

Q.4 Let S be a nonempty subset of $\mathbb{R}$. If S is a finite union of disjoint bounded intervals, then which one of the following is true?

- (A) If S is not compact, then $\sup S \notin S$ and $\inf S \notin S$
- (B) Even if $\sup S \in S$ and $\inf S \in S$, S need not be compact
- (C) If $\sup S \in S$ and $\inf S \in S$, then S is compact
- (D) Even if S is compact, it is not necessary that $\sup S \in S$ and $\inf S \in S$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Finite Dimensional Vector Spaces (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	8
2	NAT	+1	5
3	MSQ	+2	A, B, C
4	MCQ	+2	B
5	MCQ	+2	D
6	MCQ	+1	D
7	NAT	+2	3 to 3
8	NAT	+1	1011 to 1011
9	MSQ	+2	B;D
10	MCQ	+2	D
11	MCQ	+2	D
12	MCQ	+1	C

Q. No.	Type	Marks	Official Key
13	NAT	+1	60 to 60
14	MSQ	+2	C, D
15	MCQ	+2	C
16	MCQ	+1	A
17	MCQ	+1	D
18	NAT	+2	6 to 6
19	NAT	+1	4 to 4
20	MCQ	+2	C
21	MCQ	+2	D
22	MCQ	+1	C
23	NAT	+1	1.9 to 2.1
24	MCQ	+2	B

Q. No.	Type	Marks	Official Key
25	NAT	+2	1.0 to 1.0
26	MSQ	+2	A;C;D
27	MCQ	+2	C
28	MCQ	+2	B
29	MCQ	+1	B
30	NAT	+2	2
31	NAT	+1	−1
32	MSQ	+2	A;C;D
33	MCQ	+2	D
34	MCQ	+1	B

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.